(a)

GEO satellite

Pressure (hPa)

1
5
30
200
300
500
650
700
850
925

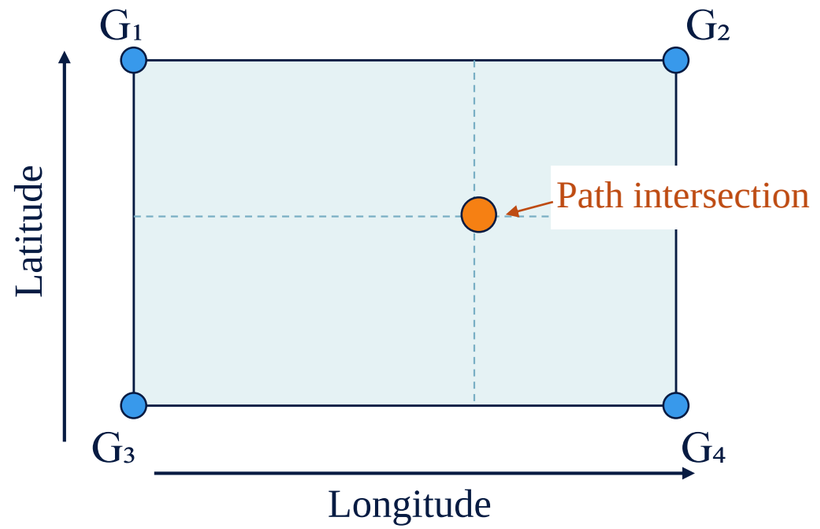
Slant propagation path

Path sampling points

Ground receiver

(b)

Local bilinear interpolation



Four neighboring ERA5 grid points

(c)

Retained meteorological inputs

50 base features

Surface quantities
Single-level diagnostics
Column-integrated quantities

106 pressure-level features

Variable-level pairs across 10 levels

50 + 106 = 156 features

Base-only candidate pool: 50

Local sampling schematic; geometry not to scale.